



Binary Quiz Kit ✕

STUDENT QUICK GUIDE

PURPOSE: LEARN BINARY
NUMBERS IN A FUN AND
INTERACTIVE WAY USING
QUIZZES AND LESSONS.

STEP BY STEP

1. **NAVIGATE MENU** - TURN THE ROTARY ENCODER TO SCROLL BETWEEN MODES. PRESS THE ENCODER BUTTON TO SELECT.

2. **QUIZ MODES** -

- BINARY → DECIMAL: CONVERT BINARY NUMBERS INTO DECIMAL.
- DECIMAL → BINARY: CONVERT DECIMAL NUMBERS INTO BINARY.
- BINARY BOMB: ANSWER QUICKLY BEFORE TIME RUNS OUT!
- LEARNING MODE: VIEW LESSONS AND EXAMPLES.

LEARNING MODE – BIT TOGGLE FUNCTION:

- EACH BIT (1 OR 0) IS SHOWN ON THE SCREEN WITH ITS VALUE (8, 4, 2, 1).
- USE THE **KEYPAD TO PRESS THE POSITION NUMBER (1-4)** TO TOGGLE THAT BIT BETWEEN 0 AND 1.
- AS YOU CHANGE BITS, THE KIT **AUTOMATICALLY CALCULATES** AND DISPLAYS THE DECIMAL VALUE.
- THIS HELPS YOU SEE HOW EACH BIT AFFECTS THE TOTAL NUMBER.

EXAMPLE:

BITS: 1 0 1 1 → DECIMAL: 11

TIPS AND REFERENCE

BINARY VALUE CHART:

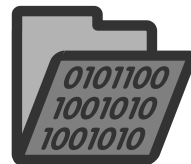
8 4 2 1

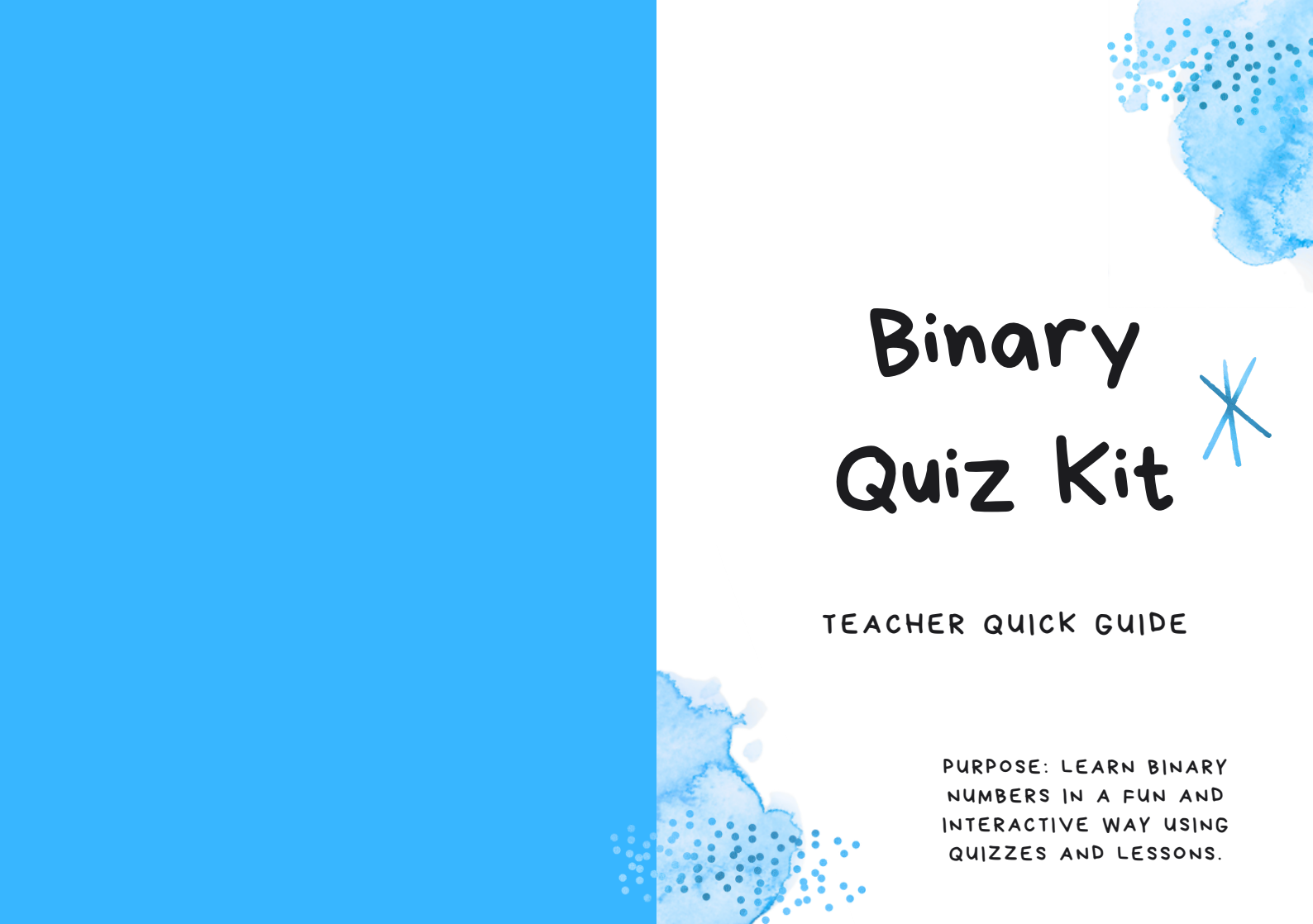
10 1 1 → 8 + 0 + 2 + 1 = 11

TIPS:

- CHECK THE QUESTION CAREFULLY BEFORE ANSWERING.
- IN BINARY BOMB, SPEED MATTERS!
- LEARNING MODE IS FOR GETTING TO KNOW BINARY BEFORE THE QUIZ.

**MESSAGE: LEARNING BINARY OPENS
THE DOOR TO UNDERSTANDING
COMPUTERS AND CODING.
HAVE FUN!**



The page features a solid blue vertical bar on the left side. In the top right and bottom left corners, there are light blue watercolor splashes containing small blue dots. The main title is written in a large, black, handwritten-style font, with a blue asterisk-like symbol to its right.

Binary Quiz Kit *

TEACHER QUICK GUIDE

PURPOSE: LEARN BINARY
NUMBERS IN A FUN AND
INTERACTIVE WAY USING
QUIZZES AND LESSONS.

SETUP & USE

BEFORE CLASS

1. **CHECK COMPONENTS** – ENSURE THE ESP32-S3, TFT DISPLAY, KEYPAD, AND AUDIO ARE WORKING.
2. **POWER ON** – SWITCH ON AND WAIT FOR THE MAIN MENU.
3. **NAVIGATE MENU** – USE THE ROTARY ENCODER TO SELECT A MODE.

USING THE KIT IN CLASS:

- **LEARNING MODE:** EXPLAIN CONCEPTS WITH PRELOADED SLIDES AND
- **QUIZ MODES:** STUDENTS CAN ANSWER QUIZES
- **BINARY BOMB:** FUN CHALLENGE FOR SPEED AND ACCURACY.

CUSTOMISATION

CHANGING QUIZ QUESTIONS:

1. **CONNECT ESP32-S3** TO YOUR COMPUTER VIA USB.
2. OPEN THE FILE **LESSON_DATA.H** IN **ARDUINO IDE**.
3. **EDIT** THE QUIZ QUESTIONS AND ANSWERS.
4. **SAVE AND RE-UPLOAD** TO THE KIT.

CHANGING MENU TITLES & SLIDES:

- EDIT MENU TITLES IN LESSON_DATA.H.
- UPDATE LEARNING MODE SLIDES BY REPLACING THE TEXT IN LESSON_DATA.H AND IMAGE DATA IN IMAGES.H.